WORKSHEET · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 2: Binary Search: Divide and Conquer

Module 1 | 45 minutes | Binary search | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. binary search ___ A. Grows very slowly as input grows (halving).
b. divide and conquer ___ B. The middle position of the current range.
c. mid ___ C. Solving a problem by splitting it.
d. logarithmic ___ D. Repeatedly halving a sorted list to find a target.

2. Plan your code

Before you build it: Binary search: nums holds 1 to 100 (sorted). Ask “Target?”. Use low, high and mid to find it; count steps. Say “Found ” target “ in ” steps “ steps”. Write or draw the blocks you will use.

3. Halving power

Each step of binary search halves the range. Roughly how many steps for 1,000,000 sorted items? Circle one.

a. A. About 20
b. B. About 1,000
c. C. About 500,000

4. Trace binary search

Searching 1–100 for 30: mid = 50, 30 is smaller so high = _____. Next mid = 25, 30 is bigger so low = _____. Binary search only works on _____ lists.

5. Explain your thinking

Compare linear and binary search. When would you choose each one? Use the words sorted and steps.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

- ☐ I can implement binary search.
- ☐ I can trace low, high and mid.
- ☐ I can compare its efficiency with linear search.